

The Nernst Equation and Cell Potential

Non-standard cells, concentration cells, E° , ΔG° and K

Name _____ Class _____ Date _____

EQUATIONS AND DATA (25 °C)

$$E = E^\circ - (0.0592 \text{ V} / n) \log Q \cdot \Delta G^\circ = -nFE^\circ \cdot \log K = nE^\circ / 0.0592 \text{ V} \cdot F = 96\,485 \text{ C/mol}$$

$$E^\circ (\text{V}): \text{Ag}^+/\text{Ag} +0.80 \cdot \text{Cu}^{2+}/\text{Cu} +0.34 \cdot \text{Zn}^{2+}/\text{Zn} -0.76$$

$$E^\circ(\text{cell}) = E^\circ(\text{cathode}) - E^\circ(\text{anode}) \cdot 1 \text{ V} \times 1 \text{ C} = 1 \text{ J}$$

Write the balanced overall equation first to find n and Q . Show units in every step.

A. Ideas

1. Explain why the voltage of a battery falls as it is used, and what the voltage is when it is completely flat.

B. Using the Nernst equation

2. For $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Zn}^{2+} + \text{Cu}$, calculate E when $[\text{Zn}^{2+}] = 0.10 \text{ M}$ and $[\text{Cu}^{2+}] = 2.0 \text{ M}$. Is E larger or smaller than E° ? Explain.

3. For $2 \text{Ag}^+ + \text{Cu} \longrightarrow 2 \text{Ag} + \text{Cu}^{2+}$: (a) calculate E° and give n (b) calculate E when $[\text{Ag}^+] = 0.010 \text{ M}$ and $[\text{Cu}^{2+}] = 1.0 \text{ M}$.

4. A concentration cell has silver electrodes in 0.0010 M and 0.10 M Ag^+ . (a) Which electrode is the anode? (b) Calculate E.

C. E° , ΔG° and K

5. For the reaction in question 3, calculate (a) ΔG° (b) K at 25 °C.

6. A one-electron reaction has $K = 1.0 \times 10^5$ at 25 °C. Calculate E° .

7. Without calculating, what can you say about ΔG° and K for a cell with a negative E° ?

D. Challenge

8. A Zn–Cu cell starts with $E = 1.10$ V. Calculate the ratio $[\text{Zn}^{2+}]/[\text{Cu}^{2+}]$ at which its voltage has fallen to 1.00 V.